

Section : Chemistry

Q.1 Which of the following formula correctly represents the double salt "Carnallite"?

1. $\text{NaCl} \cdot \text{MgCl}_2 \cdot 6\text{H}_2\text{O}$
2. $\text{KCl} \cdot \text{MgCl}_2 \cdot 6\text{H}_2\text{O}$
3. $\text{KCl} \cdot \text{CaCl}_2 \cdot 6\text{H}_2\text{O}$
4. $\text{K}_2\text{SO}_4 \cdot \text{Al}_2(\text{SO}_4)_3 \cdot 12\text{H}_2\text{O}$

Options 1. 1
2. 2
3. 3
4. 4

Q.2 The relation and units of cell constant are, respectively,

1. cell constant = l/A ; units: cm^{-1}
2. cell constant = A/l ; units: cm
3. cell constant = $l \times A$; units: cm^3
4. cell constant = l/A ; units: cm

Options 1. 1
2. 2
3. 3
4. 4

Q.3 Which of the following compounds will give Fehling's solution test?

- (A) CH_3CHO
(B) $\text{C}_6\text{H}_5\text{CHO}$
(C) $\text{CH}_3\text{CH}_2\text{CHO}$
(D) HCHO
1. (A), (B) and (D) only
 2. (A), (C) and (D) only
 3. (A), (B), (C) and (D)
 4. (B), (C) and (D) only

Options 1. 1
2. 2
3. 3
4. 4

Q.4 What is the chemical composition of Tollen's reagent?

1. Ammoniacal Silver nitrate solution.
2. Ammoniacal Sodium Nitrate Solution.
3. Ammoniacal Copper Nitrate Solution.
4. Ammoniacal Zinc Nitrate Solution.

Options 1. 1
2. 2
3. 3
4. 4

Q.5 What is the IUPAC name of Oxalic acid?

1. Butanedioic Acid
2. Propanedioic Acid
3. Ethanedioic Acid
4. Hexanedioic Acid

Options 1. 1
2. 2
3. 3
4. 4

Q.6 Match List-I with List-II

List-I List-II

Material Extra close brace or missing open brace

- (A) Gold (I) 4.5×10^3
(B) Glass (II) 3.5×10^{-5}
(C) Silicon (III) 1.0×10^{-16}
(D) Pure water (IV) 1.5×10^{-2}

Choose the correct answer from the options below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
2. (A) - (IV), (B) - (III), (C) - (I), (D) - (II)
3. (A) - (I), (B) - (III), (C) - (IV), (D) - (II)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2
3. 3
4. 4

Q.7 Which among the following is the **strongest base** in aqueous solution?

1. Methylamine
2. Dimethylamine
3. Trimethylamine
4. Aniline

- Options 1. 1
2. 2
3. 3
4. 4

Q.8 Among the following compounds, which **will not undergo oxidation reaction**?

1. Butanal
2. Butan-2-ol
3. 2-Methylbutan-2-ol
4. 3-Methylbutan-2-ol

- Options 1. 1
2. 2
3. 3
4. 4

Q.9 Match List-I with List-II

List-I

List-II

Solution

Properties

- (A) Phenol and aniline (I) Minimum boiling azeotrope
(B) Nitric acid and water (II) Maximum boiling azeotrope
(C) Ethanol and water (III) Hydrogen bonding
(D) Benzene and toluene (IV) Ideal solution

Choose the correct answer from the options below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
2. (A) - (III), (B) - (I), (C) - (II), (D) - (IV)
3. (A) - (III), (B) - (II), (C) - (I), (D) - (IV)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

- Options 1. 1
2. 2
3. 3
4. 4

Q.10 A cell is formed by combining given electrodes with $\text{Zn}^{2+}|\text{Zn(s)} (E^\circ = -0.76 \text{ V})$. Arrange the following in **decreasing order of E°_{cell}** .

Cell Notation: $\text{Zn(s)}|\text{Zn}^{2+}||\text{Given electrode}$

- (A) $\text{Ag}^+|\text{Ag(s)}: E^\circ = +0.80 \text{ V}$
(B) $\text{Na}^+|\text{Na(s)}: E^\circ = -2.71 \text{ V}$
(C) $\text{Fe}^{2+}|\text{Fe(s)}: E^\circ = -0.44 \text{ V}$
(D) $\text{Cu}^+|\text{Cu(s)}: E^\circ = +0.52 \text{ V}$

Choose the correct answer from the options given below:

1. (A), (D), (C), (B)
2. (A), (B), (C), (D)
3. (B), (C), (D), (A)
4. (C), (B), (D), (A)

- Options 1. 1
2. 2
3. 3
4. 4

Q.11 The nomenclature of the complex $[\text{CuCl}_2(\text{CH}_3\text{NH}_2)_2]$ is

1. Bis(methylamine)dichlorocopper(I)
2. Dichlorido(methylamine)copper(I)
3. Bis(methylamine)dichlorocopper(II)
4. Dichlorobis(methylamine)copper(II)

- Options 1. 1
2. 2
3. 3
4. 4

Q.12 Arrange the following transition metal ions in the **increasing order of unpaired electrons**.

- (A) Cu^{2+}
(B) Cr^{2+}
(C) Ni^{2+}
(D) Sc^{3+}

Choose the correct answer from the options given below:

1. (A), (D), (C), (B)
2. (D), (B), (A), (C)
3. (B), (A), (D), (C)
4. (D), (A), (C), (B)

Options 1. 1
2. 2
3. 3
4. 4

Q.13 Un-decomposed silver bromide from developed film, in black and white photography, makes a complex with hypo. The formula of the complex is

1. $[\text{Ag}(\text{S}_2\text{O}_3)_2]^{3-}$
2. $[\text{Ag}(\text{S}_2\text{O}_3)_2]^{2-}$
3. $[\text{Ag}_2(\text{S}_2\text{O}_8)_2]^{2-}$
4. $[\text{Ag}(\text{S}_2\text{O}_8)_2]^{3-}$

Options 1. 1
2. 2
3. 3
4. 4

Q.14 Match the metal ion given in List-I with its property given in List-II

List-I

List-II

Transition metal ion Property

- | | |
|----------------------|---------------------------|
| (A) Mn^{3+} | (I) $3d^6$, green |
| (B) Fe^{2+} | (II) $3d^4$, violet |
| (C) Ti^{4+} | (III) $3d^0$, colourless |
| (D) V^{2+} | (IV) $3d^3$, violet |

Choose the correct answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
2. (A) - (II), (B) - (I), (C) - (III), (D) - (IV)
3. (A) - (II), (B) - (I), (C) - (IV), (D) - (III)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1
2. 2
3. 3
4. 4

Q.15 Among the following, which is the correct set of reducing agents for conversion of nitrobenzene into aniline?

- (A) H_2/Pd
- (B) $\text{Sn} + \text{HCl}$
- (C) $\text{Fe} + \text{HCl}$
- (D) NaBH_4

Choose the correct answer from the options given below:

1. (A), (B), (C) and (D)
2. (A), (B) and (D) only
3. (A), (B) and (C) only
4. (B), (C) and (D) only

Options 1. 1
2. 2
3. 3
4. 4

Q.16 Which of the following reagent is **not** used to convert aldehyde into alcohol?

1. H_2/Pd
2. LiAlH_4
3. NaBH_4
4. Sn/HCl

Options 1. 1
2. 2
3. 3
4. 4

Q.17 Match List-I with List-II

List-I

List-II

- | | |
|--|--------------------|
| (A) $\text{S}_{\text{N}}1$ Reaction | (I) vic-dibromides |
| (B) Bromination of alkenes | (II) gem-dihalides |
| (C) Alkylidene halides | (III) racemisation |
| (D) Elimination of HX from alkylhalide | (IV) Saytzeff rule |

Choose the correct answer from the options given below:

1. (A) - (III), (B) - (I), (C) - (II), (D) - (IV)
2. (A) - (II), (B) - (I), (C) - (III), (D) - (IV)
3. (A) - (III), (B) - (I), (C) - (IV), (D) - (II)
4. (A) - (III), (B) - (II), (C) - (I), (D) - (IV)

Options 1. 1

2. 2
3. 3
4. 4

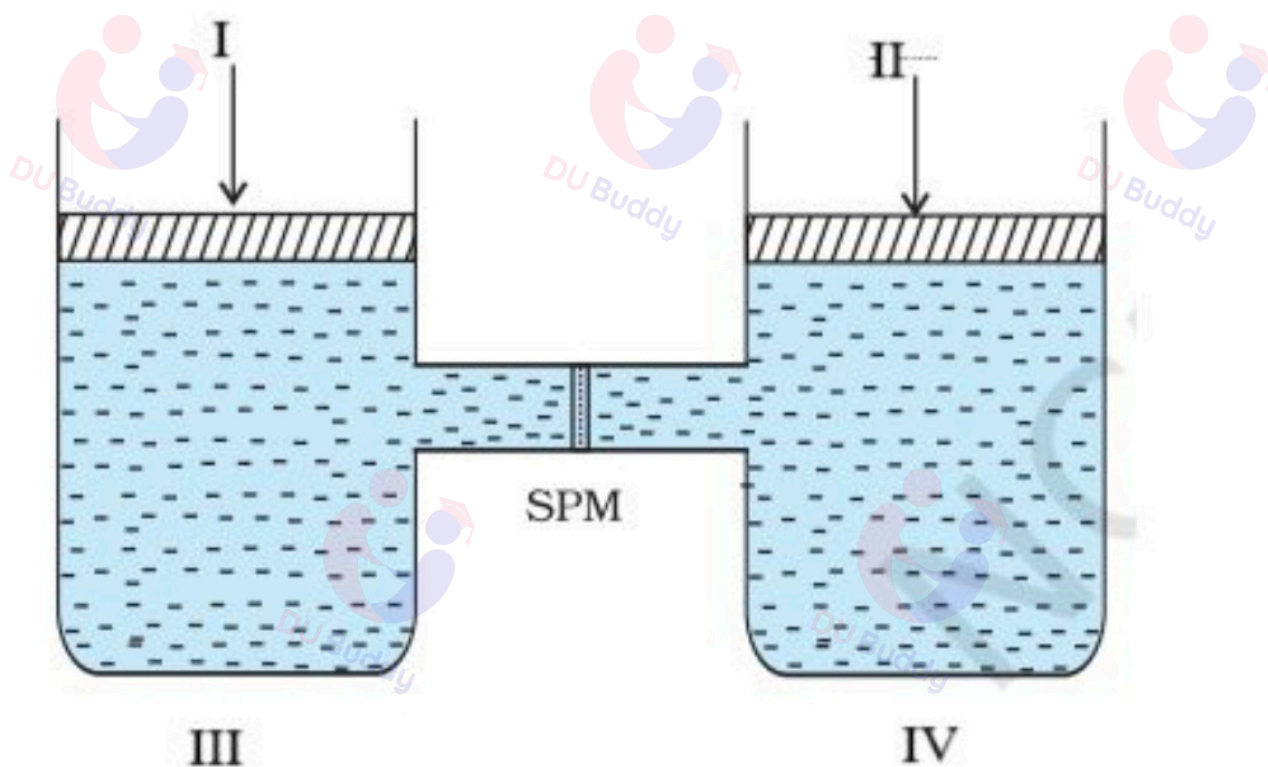
Q.18 In neutral or mild alkaline solutions, potassium permanganate oxidizes manganous salt (Mn^{2+}) to

1. MnO_2
2. Mn^{2+}
3. MnO_4^{2-}
4. MnO_4^-

Options 1. 1

2. 2
3. 3
4. 4

Q.19 In the following diagram, to prevent osmosis, if excess pressure is applied on one side of the semipermeable membrane (SPM), then I, II, III and IV are, respectively,



(A) Solution

(B) P_{atm}

(C) $P_{atm} + \Pi$

(D) Solvent

Choose the correct answer from the options given below:

1. (A), (B), (C), (D)
2. (D), (B), (C), (A)
3. (B), (A), (D), (C)
4. (C), (B), (A), (D)

Options 1. 1

2. 2
3. 3
4. 4

Q.20 Which of the following statements is correct for the galvanic cell?

1. The half-cell in which oxidation takes place is called anode and has a negative potential with respect to the solution.
2. The half-cell in which oxidation takes place is called anode and has a positive potential with respect to the solution.
3. The half-cell in which reduction takes place is called cathode and has a negative potential with respect to the solution.
4. The half-cell in which reduction takes place is called anode and has a negative potential with respect to the solution.

Options 1. 1

2. 2

3. 3

4. 4

Q.21 Which one of the following correctly represents the concentration term 'ppm'?

1. $\text{ppm} = \frac{\text{Number of parts of the component}}{\text{Total number of parts of all the components of the solution}}$
2. $\text{ppm} = \frac{\text{Number of parts of the component}}{\text{Total number of parts of all the components of the solution}} \times 10^{12}$
3. $\text{ppm} = \frac{\text{Number of parts of the component}}{\text{Total number of parts of all the components of the solution}} \times 10^6$
4. $\text{ppm} = \frac{\text{Number of parts of the component}}{\text{Total number of parts of all the components of the solution}} \times 10^3$

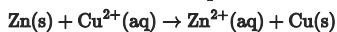
Options 1. 1

2. 2

3. 3

4. 4

Q.22 The standard electrode potential for a Daniell cell is 1.1 V. Calculate the standard Gibbs energy for the reaction:



1. $-220.27 \text{ kJ mol}^{-1}$
2. $-240.5 \text{ kJ mol}^{-1}$
3. $-212.27 \text{ kJ mol}^{-1}$
4. -198 kJ mol^{-1}

Options 1. 1

2. 2

3. 3

4. 4

Q.23 Which of the following statements is **NOT** correct regarding the catalytic ability of transition metals?

1. ability of transition metals to exist in variable oxidation state
2. decrease the concentration of reactant at catalyst surface
3. activation energy is lowered
4. weakens the bond in the reacting molecules

Options 1. 1

2. 2

3. 3

4. 4

Q.24 Complete the following analogy: S_N2 : Inversion of configuration; S_N1 :.....?

1. Racemisation
2. Retention
3. Optical rotation
4. Chiral

Options 1. 1

2. 2

3. 3

4. 4

Q.25 Which of the following complexes are **square planar**?

- (A) $[\text{Ni}(\text{CN})_4]^{2-}$
- (B) $[\text{NiCl}_4]^{2-}$
- (C) $[\text{PtCl}_4]^{2-}$
- (D) $[\text{PdCl}_4]^{2-}$

Choose the **correct** answer from the options given below

1. (B) and (D) only
2. (A), (B) and (C) only
3. (B), (C) and (D) only
4. (A), (C) and (D) only

Options 1. 1

2. 2

3. 3

4. 4

Q.26 Reaction of Phenol with conc. HNO_3 gives

1. Citric acid
2. Picric Acid
3. Nitrobenzene
4. o-Nitrobenzene

Options 1. 1

2. 2

3. 3

4. 4

Q.27 What is the reagent used in **Clemmensen reduction**?

1. NH_2NH_2
2. NaOCl
3. $\text{Zn-Hg}/\text{conc. HCl}$
4. NaHSO_3

Options

1. 1
2. 2
3. 3
4. 4

Q.28 Which among the following shows **positive deviation from Raoult's law**?

1. Ethanol and Acetone
2. Bromoethane and Chloroethane
3. *n*-Hexane and *n*-Heptane
4. Chloroform and Acetone

Options

1. 1
2. 2
3. 3
4. 4

Q.29 Nitration of aniline with **nitric acid and sulphuric acid** gives major product as

1. *m*-nitroaniline
2. *p*-nitroaniline
3. *o*-nitroaniline
4. Anilinium hydrogen sulphate

Options

1. 1
2. 2
3. 3
4. 4

Q.30 What is the product of nitration of Benzoic acid ?

1. *o*-Nitrobenzoic acid
2. *m*-Nitrobenzoic acid
3. *o*-Nitrobenzoic acid *p*-Nitrobenzoic acid
4. *p*-Nitrobenzoic acid

Options

1. 1
2. 2
3. 3
4. 4

Q.31 Which of the following is an example of vic-dihalide ?

1. Dichloromethane
2. 1,2-Dichloroethane
3. Ethylidene chloride
4. Allyl chloride

Options

1. 1
2. 2
3. 3
4. 4

Q.32 Zr and Hf (transition metals) occur together in nature and are difficult to separate. This can be attributed to

1. Filling of same *d* orbital
2. Actinoid contraction
3. Lanthanoid contraction
4. Both belong to same transition series of *d-block* elements

Options

1. 1
2. 2
3. 3
4. 4

Q.33 Identify the correct statements for **Ethylene dichloride** and **Ethylidene chloride**

- (A) They represent structural isomers.
(B) Both are formed by reaction of ethene with chlorine in presence of CCl_4
(C) Both the compounds belong to a class of dihalocompounds
(D) Both the compounds are optically inactive.

Choose the *correct* answer from the options given below:

1. (A), (C) and (D) only
2. (A), (B) and (C) only
3. (A), (B), (C) and (D)
4. (B), (C) and (D) only

Options

1. 1
2. 2
3. 3
4. 4

Q.34 On heating, $[\text{Ti}(\text{H}_2\text{O})_6]\text{Cl}_3$ becomes colorless because

- (A) Crystal field splitting is lost.
(B) $d\text{-}d$ transition occurs at higher wavelength.

(C) $3d^1$ electron of Ti^{3+} is lost.

(D) $d \rightarrow s$ transition occurs at lower wavelength.

- Options
- 1
 - 2
 - 3
 - 4

Q.35 Choose the correct order of increasing acidic strength of following compounds:

- (A) 3-Nitrophenol
(B) Phenol
(C) Propan-1-ol
(D) 3,5-Dinitrophenol

Choose the correct answer from the options given below:

1. (D), (B), (A), (C)
2. (C), (B), (A), (D)
3. (C), (A), (B), (D)
4. (C), (A), (D), (B)

- Options
- 1
 - 2
 - 3
 - 4

Q.36 What can be the product of Hoffmann bromamide degradation reaction?

- (1) $(CH_3)_2NH$
(2) $(CH_3)_3N$
(3) CH_3NH_2
(4) NH_3

- Options
- 1
 - 2
 - 3
 - 4

Q.37 Match List-I with List-II with their boiling points.

List-I	List-II
Compound name	Boiling point
(A) CH_3CH_2OH	(I) 373K
(B) CH_3OCH_3	(II) 348K
(C) H_2O	(III) 351K
(D) $CH_3CH_2OCH_2CH_3$	(IV) 248K

Choose the correct answer from the options given below:

1. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)
2. (A) - (IV), (B) - (III), (C) - (I), (D) - (II)
3. (A) - (III), (B) - (IV), (C) - (II), (D) - (I)
4. (A) - (IV), (B) - (III), (C) - (II), (D) - (I)

- Options
- 1
 - 2
 - 3
 - 4

Q.38 The correct IUPAC name for $CH_2 = CHCH_2NHCH_3$.

1. Allylmethylamine
2. 2-Amino-4-pentene
3. 4-Amino-pent-1-ene
4. N-Methylprop-2-en-1-amine

- Options
- 1
 - 2
 - 3
 - 4

Q.39 Arrange the following in order of increasing boiling points.

- (A) Bromomethane
(B) Bromoform
(C) Chloromethane
(D) Dibromomethane

Choose the correct answer from the options given below:

1. (C), (A), (D), (B)
2. (C), (B), (D), (A)
3. (D), (A), (B), (C)
4. (A), (B), (D), (C)

- Options
- 1

2. 2
3. 3
4. 4

Q.40 Which of the following statements are correct for **dilute solutions**?

- (A) The elevation in boiling point is directly proportional to the molality of the solution.
- (B) The elevation in boiling point is inversely proportional to the molality of the solution.
- (C) The depression in freezing point is directly proportional to the molality of the solution.
- (D) The depression in freezing point is inversely proportional to the molality of the solution.

Choose the correct answer from the options given below:

1. (A) and (D) only
2. (A) and (C) only
3. (B) and (C) only
4. (B) and (D) only

Options

1. 1
2. 2
3. 3
4. 4

Q.41 Read the passage carefully and answer the Questions

The term vitamin was coined from the word vital +amine since the earlier identified compounds had amino groups. Later work showed that most of them didn't contain amino groups, so the letter "e" was dropped and the term vitamin is used these days.

Vitamins are classified into two groups depending upon their solubility in water or fat. Fat soluble vitamins are soluble in fat and oils but insoluble in water. They are stored in liver and adipose (fat storing) tissues. Water-soluble vitamins are soluble in water. They must be supplied regularly in the diet as they are readily excreted in urine.

What is Xerophthalmia disease?

1. Loss of appetite
2. RBC deficient in haemoglobin
3. Bleeding gums
4. Hardening of cornea of eye.

Options

1. 1
2. 2
3. 3
4. 4

Q.42 Read the passage carefully and answer the Questions

The term vitamin was coined from the word vital +amine since the earlier identified compounds had amino groups. Later work showed that most of them didn't contain amino groups, so the letter "e" was dropped and the term vitamin is used these days.

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Which Vitamin is known as Pyridoxine ?

1. Vitamin B1
2. Vitamin B12
3. Vitamin B2
4. Vitamin B6

Options

1. 1
2. 2
3. 3
4. 4

Q.43 Read the passage carefully and answer the Questions

The term vitamin was coined from the word vital +amine since the earlier identified compounds had amino groups. Later work showed that most of them didn't contain amino groups, so the letter "e" was dropped and the term vitamin is used these days.

Vitamins are classified into two groups depending upon their solubility in water or fat. Fat soluble vitamins are soluble in fat and oils but insoluble in water. They are stored in liver and adipose (fat storing) tissues. Water-soluble vitamins are soluble in water. They must be supplied regularly in the diet as they are readily excreted in urine.

Deficiency of which vitamin causes scurvy ?

1. Vitamin C
2. Vitamin B
3. Vitamin D
4. Vitamin K

Options

1. 1
2. 2
3. 3
4. 4

Q.44 Read the passage carefully and answer the Questions

The term vitamin was coined from the word vital +amine since the earlier identified compounds had amino groups. Later work showed that most of them didn't contain amino groups, so the letter "e" was dropped and the term vitamin is used these days.

Vitamins are classified into two groups depending upon their solubility in water or fat. Fat soluble vitamins are soluble in fat and oils but insoluble in water. They are stored in liver and adipose (fat storing) tissues. Water-soluble vitamins are soluble in water. They must be supplied regularly in the diet as they are readily excreted in urine.

What are the sources of Vitamin B12?

1. Yeast, milk, egg yolk, cereals and grams
2. Meat, fish, egg and curd
3. Green leafy vegetables.
4. Citrus fruits, amla and green leafy vegetables.

Options

1. 1
2. 2
3. 3

Q.45 Read the passage carefully and answer the Questions

The term vitamin was coined from the word vital +amine since the earlier identified compounds had amino groups. Later work showed that most of them didn't contain amino groups, so the letter "e" was dropped and the term vitamin is used these days.

Vitamins are classified into two groups depending upon their solubility in water or fat. Fat soluble vitamins are soluble in fat and oils but insoluble in water. They are stored in liver and adipose (fat storing) tissues.

Water-soluble vitamins are soluble in water. They must be supplied regularly in the diet as they are readily excreted in urine.

Among the following, water-soluble vitamin is

1. Vitamin A
2. Vitamin D
3. Vitamin C
4. Vitamin K

Options 1. 1

2. 2
3. 3
4. 4

Q.46 Read the passage carefully and answer the Questions

The concentration dependence of the rate of a reaction is called the differential rate equation. It is not always convenient to determine the instantaneous rate, as it is measured by calculating the slope of the tangent at point 't' in the concentration vs time plot. This makes it difficult to determine the rate and hence the order of the reaction.

This difficulty is overcome by integrating the differential rate equation. This integrated rate equation gives a direct relation between concentrations at different times and the rate constant. The integrated rate equations are different for the reactions having different reaction orders.

The rate constant of a reaction is $2.1 \text{ mol L}^{-1}\text{s}^{-1}$. What is the order of the reaction?

1. Zero order
2. First order
3. Second order
4. Third order

Options 1. 1

2. 2
3. 3
4. 4

Q.47 Read the passage carefully and answer the Questions

The concentration dependence of the rate of a reaction is called the differential rate equation. It is not always convenient to determine the instantaneous rate, as it is measured by calculating the slope of the tangent at point 't' in the concentration vs time plot. This makes it difficult to determine the rate and hence the order of the reaction.

This difficulty is overcome by integrating the differential rate equation. This integrated rate equation gives a direct relation between concentrations at different times and the rate constant. The integrated rate equations are different for the reactions having different reaction orders.

A certain amount of N_2O_5 decomposes to half of its initial amount in 50 minutes.

If the decomposition is a first order reaction, the rate constant of the reaction is.

1. $1.38 \times 10^{-2} \text{ s}^{-1}$
2. $1.38 \times 10^{-4} \text{ s}^{-1}$
3. $2.31 \times 10^{-4} \text{ s}^{-1}$
4. $4.2.31 \times 10^{-2} \text{ min}^{-1}$

Options 1. 1

2. 2
3. 3
4. 4

Q.48 Read the passage carefully and answer the Questions

The concentration dependence of the rate of a reaction is called the differential rate equation. It is not always convenient to determine the instantaneous rate, as it is measured by calculating the slope of the tangent at point 't' in the concentration vs time plot. This makes it difficult to determine the rate and hence the order of the reaction. This difficulty is overcome by integrating the differential rate equation. This integrated rate equation gives a direct relation between concentrations at different times and the rate constant. The integrated rate equations are different for the reactions having different reaction orders.

For an integrated first order reaction, a straight line graph is plotted. The incorrect co-ordinates and resulting slope of the graph are

1. $\log[R]$ vs t ; Slope = $-k/2.303$
2. $\ln([R]_0/[R])$ vs t ; Slope = k
3. $\ln[R]$ vs t ; Slope = k
4. $\log([R]_0/[R])$ vs t ; Slope = $k/2.303$

Options 1. 1

2. 2
3. 3
4. 4

Q.49 Read the passage carefully and answer the Questions

The concentration dependence of the rate of a reaction is called the differential rate equation. It is not always convenient to determine the instantaneous rate, as it is measured by calculating the slope of the tangent at point 't' in the concentration vs time plot. This makes it difficult to determine the rate and hence the order of the reaction.

This difficulty is overcome by integrating the differential rate equation. This integrated rate equation gives a direct relation between concentrations at different times and the rate constant. The integrated rate equations are different for the reactions having different reaction orders.

A reaction is first order in A and second order in B. If the concentration of B is doubled its initial concentration and A is reduced to $3/4$ th of its initial concentration, the new rate of the reaction is

1. remains same as before
2. doubles
3. triples
4. becomes half of initial rate

Options 1. 1

2. 2
3. 3
4. 4

Q.50 Read the passage carefully and answer the Questions

The concentration dependence of the rate of a reaction is called the differential rate equation. It is not always convenient to determine the instantaneous rate, as it is measured by calculating the slope of the tangent at point 't' in the concentration vs time plot. This makes it difficult to determine the rate and hence the order of the reaction. This difficulty is overcome by integrating the differential rate equation. This integrated rate equation gives a direct relation between concentrations at different times and the rate constant. The integrated rate equations are different for the reactions having different reaction orders.

Choose the **correct statement** regarding kinetics of a chemical reaction.

1. The unit of rate constant for a second order reaction is $\text{L mol}^{-1} \text{s}^{-2}$.
2. For complex reactions, order is given by the slowest step and the molecularity of the slowest step is the same as the order of the overall reaction.
3. For zero order reaction, $t_{1/2}$ is independent of the initial concentration of the reactant.
4. Decomposition of N_2O_5 is an example of pseudo first order reaction.

Options 1. 1

2. 2

3. 3

4. 4